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M-201 STEAM TURBINE

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M 201 STEAM TURBINE

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with **Part 4, Subsection 1.2 "Summary of Work"**.

The Scope of work described hereinafter is for one (1) Steam Turbine which has a capacity in the range of 150 MW-300 MW at site condition (new and clean machine), for a Combined Cycle Power Plant per Block. The total number of Steam Turbine for three (3) Blocks (Block 2, 3 & 4) will be three (3) Steam Turbines.

The Steam Turbine and its auxiliary systems shall be according to manufacturers proven standards. The scope specified below refers to a double **or triple** pressure cycle. In such case the scope shall be adjusted correspondingly, to the approval of the Owner.

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The Steam Turbine shall be complete with all accessories and appurtenances required for a complete functional system and as hereinafter specified, including but not limited to:

- a. Steam Turbine.
- b. HP steam stop valves and control valves
- c. **IP (if applicable)** & LP steam stop valves and intercept valves
- d. Permanent strainers with fine orifices
- e. Strainer for start-up
- f. Turbine exhaust explosion diaphragms
- g. Exhaust hood sprays
- h. Turbine oil system, including jacking pump (if required)
- i. Gland steam sealing system
- j. Turbine hydraulic fluid system
- k. Turning gear
- l. Electro hydraulic control system (EHC), and turbine supervisory instrumentation

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- m. Sole plates, sub sole plates and final grouting
- n. Foundation for all equipment specified in this Specification
- o. All electrical and I&C equipment required for the Steam Turbine.
- p. Vibration measurement and monitoring equipment for each bearing of the main shaft of the Steam Turbine Generator Set, including the necessary processing and display equipment in the Central Control Room.
- q. Turbine / Generator enclosure as necessary to maintain the required noise level.
- r. The scope of work shall include, but not limited to, the following works and equipment for the proper performance of steam turbine generator complete with all auxiliary plants equipment:
 - All necessary pipe work and fittings expansion joints, valves, automatic actuators, safety devices and fastenings etc. for the systems mentioned above.
 - Necessary drain tanks, interconnection piping instruments and controls etc.
 - Motor operated drain valves with drain piping, hangers and orifices etc.
 - All necessary venting, draining and emptying equipment, complete
 - All necessary coupling and covers
 - All necessary safety measure

2. **CODES AND STANDARDS**

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5.**

3. **DESIGN AND OPERATING CONDITIONS**

The rated output-of the steam turbine unit shall be in accordance with the Combined Cycle Power Plant total output requirement. The heat cycle design shall be in line with **Part 4 Section 3, G 001.**

The Steam Turbine shall be of standard proven design.

The Bidder shall include in his Proposal a formula and chart relating the anticipated running hours between overhauls in relation to the number of starts and

operating load pattern. Anticipated minor and major overhauls, routine maintenance intervals, and with the nature and duration of the work to be undertaken shall all be indicated on the chart. The auxiliary equipment supplied with the steam turbine should also be included.

The Bidder shall ensure the design of the steam turbine generator offered is completely suitable for the design conditions identified in this Specification including all the operating and loading regimes identified.

The Bidder shall define the variable pressure characteristics of the steam turbine including the minimum load at which variable pressure operation can be sustained.

Note shall be taken in the design of AC essential auxiliary drives of the philosophy described in electrical specifications providing a secure essential service.

The Bidder shall design the steam turbine proper and its auxiliaries using good engineering practice and proven experience, in order to obtain good and reliable equipment having superior performance. No prototype equipment or components shall be used.

Further following Design requirements shall be considered:

Performance and Design Criteria	Unit	Data
Operation mode	-	Constant or variable pressure
Permissible deviations of live steam pressure and live steam temperature	-	According to IEC recommendations, latest edition.
Absorption capacity (throughput ref. to maximum continuous rating of turbine generator) at 96% nominal live steam pressure (constant pressure mode).	%	100
Maximum continuous rating of steam turbine generator is defined at maximum Gas Turbine base load under Guarantee Conditions specified in Part 4 subsection 2.6).		
Absorption capacity at increased nominal live steam pressure (variable pressure mode)	%	105

Performance and Design Criteria	Unit	Data
Permissible tolerance of absorption capacity	%	-0/+4
Permissible frequency deviations at maximum continuous rating:		
• without limitations	%	-3 to +3
• for maximal 20 minutes duration but not exceeding 2 hours per year	%	-4
• for maximal 10 minutes duration but not exceeding 1 hour per year	%	-5
Following operating cases are to be considered as minimum:		Refer to Part 4 Section 3, G-001
Operation time	EOH	200000
Max. speed overshoot in case of full load rejection	%	8

Guarantees and Criteria for Rejection	Unit	Data
1 General Guarantees		
Sensitivity of electro-hydraulic speed governing system including governor valves	%	less 0.06
Maximum increase in running speed in case of sudden total or partial load reduction	%	8
Vibration		Refer to Part 4 Section 3, M-014
Maximum noise pressure level at 1 m distance from the noise source (without other external noise sources)	dB(A)	Refer to Part 4 Subsections 2.9 and 2.16
Construction tolerances according to the relevant regulations must have already been accounted for in the guarantee values.		

Guarantees and Criteria for Rejection	Unit	Data
2 Performance Guarantees		Refer to Part 4, Subsection 2.6 and Part 5 Schedule 6
3 Criteria for rejection		
Should the deterioration of the guarantee values be greater than following values then the Owner shall have the right to reject the scope of this section in its entirety		
Gross power output at generator terminals at MCR (related to GT Base load)	%	5
Starting and loading time (not applicable)	% (not applicable)	30 (not applicable)
Vibration		refer to Part 4 Section 3, M014

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4. DETAILED REQUIREMENTS

4.1 Arrangement

The turbine generator set and its auxiliaries shall be located in the Steam Turbine Building as shown in the attached Bid Drawings. The Bidder shall provide sufficient laydown area inside the Steam Turbine Building to perform a major overhaul of the steam turbine and its auxiliaries and providing easy access for the operation and maintenance work. The Bidder shall submit with his Proposal a laydown drawing showing the area needed for a major overhaul where turbine and generator will be dismantled.

The Bidder shall provide all platforms, galleries and stairways necessary to give access to the various sections of the plant being supplied under this Contract. They shall provide adequate means of access for all operation, inspection and overhaul purposes and shall be of sufficient strength to support workmen, tools and portions of plant which may be placed on them during overhaul and inspection periods. Ladders are only to be provided as an extra means of escape.

If during evaluation or after award of Contract the laydown area and the access for operation and maintenance is considered not to be sufficient, the Bidder shall improve his Proposal by adjustment of the building size without price adjustment.

4.2 Turbine Casings

The Bidder shall provide details of the system of supporting casing and accommodating expansions.

All major connections to the turbine casing shall be welded; loop pipes shall be flanged at suitable points to permit removal for maintenance and inspection purposes. All necessary lifting gear (including guide rods if required) shall be included.

Bolt heating or any other heating equipment or other special equipment or devices necessary to permit dismantling and reassembly of casing shall be supplied by the Bidder. Where such equipment is electrically operated, details of voltage and rating shall be included in the Proposal. After completion, the equipment shall be left with the Owner in a good operating condition.

The cylinder shall be provided with relief diaphragms arranged to discharge vertically into the turbine hall. Deflectors, where necessary, shall be provided to direct any steam away from personnel or operating areas. The cylinder shall be designed to minimize the effects of moisture erosion and corrosion.

4.3 Turbine Rotors and Couplings

The completed turbine rotors with blading shall be statically and dynamically balanced in the shop and an overspeed test shall be performed.

Critical speeds for each rotor shall be outside the limits of 10 % below and 15 % above rated speed.

Couplings shall be designed to allow for removal of the turbine rotors without opening the generator.

All rotors with shrunk on and keyed discs are not acceptable.

4.4 Fixed and Moving Blading

Moving blades and fixed blades or nozzles shall be of proven design in all respects.

LP rotor moving blades in the wet region shall be suitably protected against erosion and a detailed description of the method used to achieve this shall be included in the proposed.

4.4.1 Bolts

Bolts subject to high temperature shall be made of suitable high temperature resistant material.

4.4.2 Thermal Insulation

The Bidder shall include for the supply and application of all necessary thermal insulation required for the Works, and it shall comply in full with **Part 4 Section 3, M-003, “Thermal Insulation and Lagging”**.

4.5 Steam Turbine Shaft Glands and Gland Sealing System

4.5.1 Shaft Glands

Metallic labyrinth steam sealed shaft glands shall be provided.

All diaphragm and shaft glands shall be designed to prevent damage if rubbing occurs. Reliability is of the utmost importance.

4.5.2 Gland Sealing System

The gland sealing system shall be capable of maintaining full condenser vacuum with the turbine on barring gear and during start-up and shut down, in addition to normal operation, and shall be self regulating under all conditions

The gland steam sealing system shall be automatic and supplied with the gland steam piping which shall terminate at the main and if required at the auxiliary steam piping. Suitable nozzles for steam supply connections on the main steam piping and auxiliary steam header shall be provided

4.5.3 Gland Steam Exhauster System

A gland steam condenser, two 100 % capacity motor driven exhauster fans (or steam operating ejectors) and all piping, valves, traps, controls, instrumentation and panels shall be supplied. Tube material of gland steam condenser shall be stainless steel.

Air exhaust pipes, relief valve vent, etc shall be carried outside the turbine room and shall have weather protection, bird screens and rain hoods, as required.

The gland steam condenser shall drain to the plant main condenser.

4.6 Steam Turbine Governing and Protection Philosophy

4.6.1 General

An electrohydraulic governor shall be provided for turbine load and frequency regulation during normal on-load operation and turbine run-up, and shall be fully compatible for operation in the combined cycle mode with the gas turbine and waste heat boiler equipment. The governor system shall be able to operate in fixed pressure, sliding pressure and modified sliding pressure mode.

The design requirements for the electronic equipment with the governor and protection system are set out under The Control and Instrumentation

Specification **Part 4 Section 3, I-100, “Plant Control System”**.

A detailed description of the governor and protection system shall be included in the Proposal

4.6.2 Turbine Governors

4.6.2.1 On -Load Governor Control

The governor shall be subject to the following requirements:

- a. It shall be a speed governor with adjustable droop characteristic including an infinite droop setting.
- b. The governor shall be capable of pressure governing either with or without frequency droop influence.
- c. The response shall be sufficiently rapid and incorporate any necessary derivative and anticipatory functions or speed limiting devices to ensure an

adequate speed margin between the turbine speed rise, following a full load rejection, and the operating point of the overspeed emergency governor, i.e. in case the steam turbine generator set is separated from the electrical grid when operating at any load up to full load, the set shall not trip but maintain no load operation. All other parts of the works shall be designed accordingly. Speed control facilities shall be provided to allow the controlled overspeeding of the turbine if required on a regular basis for testing of the overspeed emergency governor .

The Bidder shall provide a control block diagram and transfer function identifying the response characteristic of the governor and the influence of any adjustment provided.

4.6.2.2 Start-Up Governor Control

The full range speed governor or a separate start-up speed governor shall control the turbine from barring speed to the over-speed level. The governor is to be used in conjunction with the automatic run up control system which shall generate the speed reference signal to a wide range speed governor. It shall also be possible to carry out a controlled overspeed test if such is required on a regular basis.

4.6.2.3 Overspeed Governor

- a. Two entirely separate or a 2 out of 3 redundant governor system shall be provided. They shall be set to operate at a speed above the rated speed as specified in **Part 4 Section 3, I-100**. Each system shall be capable of being independently tested both on-load by simulation and also off-load by raising the actual shaft speed to overspeed trip levels. Interlocks shall be provided to ensure that both systems cannot be tested simultaneously, one system always remaining in service, and that the tripping system cannot be reset from its test to its normal position before the overspeed trip ring (or equivalent) has returned to its normal position to avoid inadvertent tripping of the unit.
- b. The on-load testing of the overspeed governor channels shall be carried out remotely from the Main Control Room, as shall all off-load actual over-speed testing of the turbine if such is required on a regular basis. All necessary control initiating devices and indicators, including a digital turbine speed display, required in the Main Control Room shall be supplied.

- c. A unit trip due to the operation of either channel of the overspeed governor shall require manual re-setting of the trip mechanism. A hand reset facility for the trip mechanism shall be provided as the remote manual reset facility in the Main Control Room, required for the normal reset function.
- d. The additional speed rise after an overspeed trip at full load, following the failure of the normal governing devices shall have an adequate margin below the 120% overspeed
- e. In case of mechanical devices facilities for ready access to adjust the trip setting shall be provided with only the removal of access covers.
- f. Any restrictions on routine overspeed testing of the turbine generator shall be detailed in the Proposal.

4.6.3 Turbine Protection

To ensure the turbine is protected from a hazardous operating condition the steam turbine shall trip or be unloaded when conditions which threaten the integrity of the plant are detected. Any trip or unscheduled unloading shall be accompanied by an audible and visual alarm which shall initiate movement of the gas turbine exhaust dampers to the bypass position. The tripping equipment shall operate independently of any turbine governing or automation equipment. The equipment shall work in conjunction with the generator protection equipment and circuit breaker tripping relays.

4.6.3.1 Turbine Tripping

The arrangements of the turbine tripping system shall comply with the requirements set out under Control and Instrumentation Specification.

The following conditions shall initiate a turbine trip

- a. Overspeed
- b. Remote and local trip button initiation
- c. Low lubricating oil level
- d. Low lubricating oil pressure
- e. High bearing oil temperature
- f. Thrust bearing failure
- g. High condenser back pressure

- h. Overheat of LP exhaust end
- i. Generator trip
- j. High shaft and or pedestal vibration
- k. Boiler pressure too low

4.6.3.2 Turbine Unloading

Progressive turbine unloading shall be provided for falling boiler pressure when below the variable pressure set point level to protect the turbine against water carry over.

4.6.3.3 Turbine Bypass System

A fully integrated steam bypass system shall be installed as specified under **Part 4 Section 3, M-302, "Turbine Bypass System"** for the following purposes:

- a. Rapid matching of steam and turbine metal temperature during start-up of the turbine in order to reduce the time of the start-up
- b. To provide the facility by which the boilers can continue to generate steam, albeit at a reduced rate, when load is either temporarily lost on the steam turbine generator or is rapidly reduced.
- c. As a pressure relief system to reduce the risk of lifting the boiler safety valves with instances of rapid load reduction on the steam turbine.

4.6.3.4 Manual Trip

A manual tripping device shall be provided in the Main Control Room and local to the steam turbine pedestal.

4.6.3.5 Exhaust Hood Cooling

- a. The turbine exhaust hood shall be protected against over temperature due to last blade windaging when operating the steam turbine at low loads for extended periods.
- b. The load range over which the cooling system would be required to prevent over heating of the hood shall be stated.

4.7 Emergency Stop and Governing Valves

4.7.1 Arrangement of Valves

4.7.1.1 HP Cylinder Steam Supply

- a. Two or more isolating and emergency stop valves shall be provided for controlling the inlet of main steam to the H.P. turbine section.
- b. Sufficient governing valves shall be provided at the inlet of the H.P. cylinder to fulfill all requirements of the governing system.
- c. Supervisory thermocouples shall be provided at recommended points to indicate metal and steam temperature and to enable thermal stress levels to be evaluated. The requirements for such thermocouples are detailed in the Control and Instrumentation Requirements **Part 4 Section 3, I-001 “Instrumentation and Control”**.
- d. A power operated drain (or parts of the turbine drains system) shall be arranged to drain and/or allow the flow of warming steam from each emergency stop valve chest just upstream of the valve seat. This shall have an adequate steam flow capacity to facilitate warming through the steam mains at start-up. If that start-up control of the turbine is achieved by the governing valves, this warming drain or vent facility shall be arranged to be connected as appropriate just upstream of each group of governing valves.

4.7.1.2 IP (if applicable) & LP Cylinder Steam Supply

- a. The **Intermediate** pressure steam admitted to the **I.P.** cylinder shall be supplied from the H.P. cylinder and the **I.P.** steam manifold. The latter steam supply may be admitted through a single emergency stop valve and a single control valve which may be incorporated into one steam chest. **The low pressure steam admitted to L.P cylinder shall be supplied from the I.P cylinder**
- b. All the technical and operating requirements described for the H.P. section inlet valves shall apply.
- c. If a single control valve is provided for admission of the **Intermediate** pressure steam, then on load testing shall be achieved in conjunction with the **I.P.** bypass system to enable the valves to be provided without risk of lifting the **I.P.** safety valves on the HRSG.

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4.7.2 Control Design Requirements.

The valves, spindles, guide bushings, and glands shall be of compatible materials entirely suitable for the operating conditions. Valve seats and valve plugs or heads shall be hardened or provided with hard alloy areas to minimize erosion. Spindles, guide bushings and glands shall operate for prolonged periods of operation without pick-up, seizure or permanent change of dimensions.

Full details of the materials proposed for the valves, seats, spindles and guides, shall be provided in the proposal along with details of operating experience with these materials.

The H.P. turbine steam valves shall be designed to allow full-arc admission under all operating conditions and particularly during run-up of the machine and when operating the steam plant in sliding pressure mode

If the valves offered are subject to valve spindle steam leakage with turbine out of service, an estimate of leakage steam under full boiler pressure conditions shall be given. The Bidder shall also give the location of the leakage and drain destinations of each spindle leak-off. All steam leakage from the valve spindles shall be recovered.

The servomotors of all turbine emergency stop and governing valves shall be operated hydraulically by oil supplied from the governing oil system. All these valves shall be arranged to be held open by hydraulic pressure and to be closed (against full steam pressure) by spring action.

All necessary equipment and design features shall be provided to ensure the rapid closure of all valves in the event of turbine trip.

Provision shall be made for remote operation (tripping and re-setting) of the emergency stop and governing valves. These valves shall all close automatically on loss of governing oil pressure. Adequate provision shall also be included to facilitate testing each valve and servo for satisfactory operation with the turbine either off- load or on-load.

The Bidder shall describe fully the arrangement and methods of operation proposed for the emergency valves and the protection provided to prevent the turbine speed reaching a dangerous value in the event of the normal governing equipment failing to function on loss of load.

Hydraulic power and control equipment for steam valves shall have a positive means of lubrication for all sliding and rotating bearing mechanism. Where these mechanism are exposed to atmospheric conditions they shall be adequately protected by the dust covers or seals to prevent dust build-up in bearings and on journals, plungers or similar small clearance parts. Suitable venting shall be provided for all servomotor enclosures to ensure water vapour is removed to prevent corrosion of enclosed elements.

4.7.2.1 Emergency Stop Valves

- a. In addition to the aforementioned requirements for Emergency Stop and Governing Valves, the H.P. emergency stop valve shall incorporate an internal pilot valve, if necessary, to enable the main valve to be opened against full steam pressure and for obtaining sufficiently fine control for starting up the set (when not using a governing valve controlled method).
- b. The H.P. and I.P. emergency stop valves shall have facilities for back-seating when in the fully open position.
- c. The emergency valves shall be provided with all necessary limit switches or position transducers for turbine tripping and for the electrical protection equipment. The limit switches for the electrical protection equipment shall be arranged to prevent emergency tripping of the generator main circuit breaker and field suppression switch until and unless the emergency valves are fully closed.

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4.7.2.2 Governing Valves

- a. The design of the governing valves shall make provision for reducing the thermal effects of sudden temperature variations associated with the operation of the valve and, if necessary, arrangements shall be incorporated to maintain the steam passage.
- b. Governing valves shall incorporate a back seating arrangement to minimize steam leakage and foreign matter gaining access to the clearance space between the spindle and its bushing or packing.
- c. Where a nozzle governed turbine is offered, facilities shall be included to enable the governor valve operating sequence to be varied to suit both fixed pressure operation and variable pressure operation. These facilities shall comprise arrangements which allow the governing valves to be operated

sequentially or simultaneously, or alternatively to be operated partly sequentially and partly simultaneously as appropriate. It is preferred that these facilities be available on-load via the turbine governor

- d. Turbine governing valves shall be designed to be free of vibration and steam buffeting under all operating conditions from start-up to full load, It is preferred that governing valve design is such that a null loading situation (i.e. zero nett steam force on valve head) is not possible under any conditions of steam pressure and turbine load. Should the design offered not comply with this preference, the Bidder shall detail those design features incorporated to avoid valve head vibration when operating at or near the null point, and shall provide details of operational experience with the design.

4.7.3 On-Loading Testing

Facilities shall be provided for the regular on-load testing of each stop and emergency valve, and each governing valve.

The Bidder shall describe the provisions made and the procedure to be adopted when on-load testing these valves. The test facilities shall be located in a such position that one operator can select, test and confirm the correct operation of each valve in turn. The Bidder shall state what reduction in load he expects will necessary to allow, this testing to be carried out. Any provisions required in the way of cross-over piping to minimise loss of load under on- load test conditions shall be included and described in the Proposal.

4.8 Steam Strainers

- a. Permanent steam strainers shall be provided in the H.P. **IP** and L.P. steam circuit to prevent the entry into the machine of any material which could damage blading or valve seats. Each strainer may be a separate item or an integral part of the respective emergency valve and shall be arranged to facilitate cleaning.
- b. After the initial commissioning period the steam strainers shall be checked. In case of a damage the Bidder shall replace the strainers without additional cost.
- c. Differential pressure gauges shall be provided to give local indication of the differential pressure across each temporary and permanent steam

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strainer. Pulsation damping devices shall be provided with the differential pressure gauges.

4.9 Turbine and Pipework System Drains

- a. All parts of the turbine and steam pipework where water is liable to collect shall be drained and wherever possible any clean drainage shall be recovered and returned to the condensate system.
- b. The steam and condensate drain system provided shall be complete from the point of drainage to the final disposal points. They shall provide for the safe and efficient start-up and operation of the turbine generator unit, controlled remotely and automated as described elsewhere in the specification.
- c. No steam drains shall be directed into hot piping.
- d. Internal drains on the turbine shall be provided to ensure that water will not collect in the casings, particularly between the L.P. section exhaust end diaphragms.
- e. Bidder shall fully describe his proposed drainage schemes, and shall include particular reference to these considerations.
- f. Drain vessel(s) shall be complete with inspection access arrangements and automatic desuperheating sprays as necessary to prevent overheating of the condenser and turbine exhaust under all conditions of start-up and operation.
- g. Drainage from the turbine cylinders, Steam chests, loop pipes and associated bleed steam pipes shall be controlled by duplicated motorized/pneumatically operated isolating valves mounted in series.
- h. All important turbine drains including all superheated drains shall be provided with duplex thermocouples for remote monitoring.
- i. Remote monitoring of drains temperature shall be provided and arranged to be displayed as appropriate in conjunction with the corresponding saturation temperature derived from the steam pressure existing at the point of drainage.

4.10 Barring Gear

- a. The Bidder shall provide power operated barring gear for the safe and reliable operation of the turbine generator. The equipment shall be complete

and shall be fully automatic in operation, provision shall be made for manual initiation of the power operated barring gear, engaging and disengaging as necessary during the start-up and shut down sequence.

- b. Lubricating oil pumps shall be adequately interlocked with the barring gear to prevent rotation of the rotor without lubricating oil at the appropriate pressure.
- c. The barring system shall be driven by AC motors.
- d. Facilities shall also be provided for manually turning the turbine generator rotor, with the jacking pump in operation.
- e. The period of barring before start-up and after run down under hot and cold conditions shall be stated in the Proposal, and the reasons given for such a period. Full details of the barring system shall be included in the Proposal.

4.11 Bearings

- a. Bearings shall be of the pressure lubricated journal type and shall be accessible without requiring removal of turbine casings. They shall be of the horizontally split and self-aligning type.
- b. The thrust bearing shall be of Michell or Kingsbury tilting pad type and shall be capable of handling the maximum possible thrust in either direction under any combination of operating conditions. A thrust bearing wear indicator shall be provided with an alarm for excessive thrust wear.
- c. Each bearing housing shall be fitted with an oil baffle on the side adjacent to the cylinder glands to prevent oil leaving and carbon deposits forming on the oil wiper rings.
- d. A resistance temperature detector and a separate locally mounted temperature indicator shall be provided in each main bearing oil drain. In addition, thermocouples embedded in the white metal of the turbine generator bearings shall be supplied, together local and remote indication and remote alarm annunciation.
- e. The maximum permissible vibrations shall be in line with the requirements of **Clause 4.16** below.

4.12 Lubrication and Control Oil System

4.12.1 General

A complete lubricating oil system to the approval of the Owner shall be provided for each turbine generator Unit. The lubricating oil system can be fully integrated with any associated relay oil, governing oil, however also a separate relay and governing oil system would be acceptable. The systems shall have sufficient capacity to accommodate the most severe requirements.

Proximity of the oil system to high temperature steam pipes shall be avoided to minimize the fire hazard, with oil pipes shall be located below the steam pipes, wherever possible. Compression ring joints shall not be used in the oil systems. Guards and drip trays shall be provided where necessary to prevent contact between oil escaping and hot piping and other hot parts of the machine. All oil piping shall be prefabricated, cleaned and pickled to remove all mill scale, dirt, rust and extraneous material.

The systems shall include sufficient standby equipment to allow cleaning of filters and coolers without restricting in any way the output of the steam turbine unit.

The capacity of the system and duty of the main oil pumps shall be such that the number of complete changes per hour does not exceed eight.

A common lubrication and control oil system shall comprise at least the following equipment:

- a. Main lubricating oil pump
- b. Standby/Start-up/Rundown full duty auxiliary lubricating oil pump AC rotor driven.
- c. Emergency rundown lubricating oil pump DC motor driven
- d. Jacking oil pump AC motor driven and if necessary an emergency jacking oil pump DC motor driven
- e. Two (2) 100 % AC motor driven control oil pumps
- f. Duplicate 100 % capacity oil filters for each system
- g. Common oil coolers or separate for each system 2 x 100%

- h. Oil storage and settling tank complete with vents and vapour extraction fans.
- i. Oil purifier - AC motor driven, permanently connected in the lubricating oil system
- j. All necessary pipework, valves, fittings, strainers.
- k. All necessary control, interlocks and instrumentation.
- l. AC flushing lubricating oil pump. (If required)
- m. Unit lube-oil transfer pump (if required)

All bearing drain lines and oil wells are to be provided with visual indicators capable of being observed from a local platform or operating floor level

In the event of AC failure, the emergency DC operated emergency rundown lubricating oil pump should automatically come into operation. A combined AC/DC motor-driven pump with tandem motors will not be accepted.

4.12.2 Lubricating Oil Pumps

The pumps shall comply with **Part 4 Section 3, M-007, “Vertical Centrifugal Pumps”**. Starting and stopping of the motor driven pumps shall be normally automatic, but manual override facilities shall be incorporated.

The lubricating oil pumps shall be of the centrifugal or gear type. The motor driven pumps shall be located direct on oil storage tanks.

Interlocks shall be provided to prevent starting of the turbo-generator units in the absence of lubricating oil pressure.

4.12.2.1 Main Oil Pump

Suitable oil booster equipment shall be supplied if necessary to ensure that satisfactory section conditions prevail at the main oil pump under all start-up and operating conditions.

4.12.2.2 Auxiliary Oil Pump

The auxiliary oil pump shall be AC motor driven to provide full oil requirement for start-up, shutdown and when the main oil pump may be out of service for a

prolonged period. It shall provide automatic back up for main pump and start on falling oil pressure.

4.12.2.3 Emergency Rundown Oil Pump

An emergency rundown oil pump shall be provided, for the supply of sufficient lubricating oil to supply the bearings when the unit is running down after a trip, or being barred. The pump shall be DC motor driven, and shall be arranged so that it comes into operation automatically on failure of AC supplies.

The DC oil pump shall be arranged to enable the turbine- generator bearings to be fed with oil under mishap conditions such as the total loss of AC supplies.

4.12.2.4 AC Flushing Oil Pump (If required)

One 100% duty flushing oil pump AC motor driven may be provided. The pump shall supply sufficient lubricating oil to the bearings when the unit is barred or for flushing purposes.

4.12.3 Tanks

The lubricating oil storage tank provided shall be fitted with strainers which can be removed while the turbine is running, electric motor operated vapour extractor (s) discharging outside the building, oil level indicator, low alarm initiating device, manhole with cover and all accessories.

The tank top shall be arranged as an oil tray with drain connection. The tank shall be blasted and thoroughly cleaned before priming and painting. Refer to **Part 4 Section 3, M-012.**

Temporary magnetic type oil filters shall be provided for initial operation.

Oil sight boxes shall be provided in all pipes returning oil by gravity from each turbine generator bearing to the oil tank.

4.12.4 Filter

Duplicate 100% capacity filters shall be provided for the feed to each system. Changeover from one element to the other shall be possible without reduction in flow or a momentary droop in pressure. The changeover valves shall be interlocked or arranged so that it is not possible to inadvertently cut off the supply of oil to the plant. A differential pressure gauge shall be provided across the

filter unit. Adequate provision shall be made for draining and venting the section of filter undergoing maintenance.

The degree of filtration shall be of the coarsest possible consistent with satisfactory operation and wear life of the turbine- generator bearings, gears and other lubricated items.

4.12.5 Oil Coolers

Two 100% capacity oil coolers shall be supplied either for common system or separate for each system. The oil in the coolers shall be cooled by demineralised water.

The heat exchangers shall be designed with sufficient margin to allow for fouling in accordance with T.E.M.A. Standards. If necessary, control shall be provided at the oil coolers to maintain automatically the required oil temperature leaving the coolers. The oil pressure shall be higher than the water pressure.

4.12.6 Lubricating Oil Purifier

An AC motor driven oil purifier, for water and sludge removal from the lubricating oil, shall be provided with an hourly capacity of not less than 20 % of the total oil quantity in the lubricating and relay systems. It shall be permanently connected in a bypass to the lubricating oil system in such a manner that it can be operated when the turbine generator is running or shut down.

The purifier shall be fully self-contained on an oil-tight baseplate with automatic lubrication as necessary. It shall be provided with all necessary pipe connections, including permanent connections for drainage to the dirty oil drains system and for supply of the sealing water to be taken from the feedwater make-up system.

Those parts of the purifier that are liable to corrosion, in particular the bowls, shall be manufactured from stainless steel

In the event that the purifier can pass lubricating oil to waste, then automatic shutdown and alarm facilities shall be provided.

The purifier shall be provided with an indirect electrical heating system with thermostatic control arranged to heat the incoming oil as necessary for optimum purifier performance. This heating system should also provide for increased heating capacity to be utilized in conjunction with elevated oil temperature for

oil flushing purposes, both during initial commissioning and subsequently after each unit overhaul.

4.12.7 Lubricating Oil Pipework and Valves

The pipework and valves shall comply with **Part 4 Section 3, M-001 and M-004**. Provision shall be made for the collection and detection of any leaking oil. Details of the oil leakage guard system shall be given in the Proposal.

Protection against overpressure in the lines shall be provided. Relief lines from pump discharge piping shall be directed back to the supply tank.

4.12.8 Dirty Oil Drain System

If required one Dirty Oil Drains Collection Tank and all associated pipework shall be provided to collect turbine oil by gravity drainage from all oil leakage, venting, and draining points such as pump glands, control valves, oil strainers, drip trays, vent and drain tundishes and oil purifier. The Dirty Oil Drains Collection Tank shall be pumped / drained to the Dirty Oil Tank to be supplied in line with **Part 4 Section 3, M-209**. The pump and pipework shall be permanently installed.

The materials and procedure of construction of the tank and system shall comply fully with the requirements of the lubricating oil system. The drains tank shall be located within a bunded area. A tank drain shall be provided, suitably valved and fitted with a blank flange. The drain pipe shall be suitable for use as a puddle pipe where it passes through the bund

4.13 Shaft Drive Couplings and Alignment

The couplings shall be designed for the worst anticipated conditions of a short circuit reaction from the generator or an attempted out-of-phase synchronization. Details of the coupling design basis and of any gear box shall be stated in the Proposal.

4.14 Miscellaneous Requirements

a. Materials

All materials not specifically covered herein shall be of the best quality for the service contemplated. All materials shall conform to the ASTM or

equivalent Standard Specifications. No test specimens will ordinarily be required over and above the mandatory requirements applicable to the ASTM Standard Specifications

b. Guards

Safety guards shall be provided for all rotating shafts and shaft couplings and for other exposed moving parts of the complete turbine-generator unit. All guards shall be in accordance with applicable OSHA standards

c. Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter, and shall be in accordance with **Part 4 Section 3, M-001, S-420 and S-430.**

d. Protective Coatings

All exposed, machined surfaces shall be coated with a suitable grease or antirust compound -before shipment. Inlets and outlets shall be blanked off or plugged before shipment.

4.15 Turbine / Generator Enclosure

Turbine-generator appearance cover shall be provided in accordance with the Bidder's standard design and also where required for noise attenuation purposes.

Where fitted, the appearance cover shall be of pleasing appearance, suitably stiffened and secured to avoid thermal distortion. It shall be resistant to corrosion and mechanical damage, and shall have adequate provisions to allow free access to all areas of the steam turbine and generator for maintenance and operation.

For the purpose of reducing maintenance outage periods to a practicable minimum, the appearance cover shall be designed to be erected in sections, each section being readily removable. The sections shall be strengthened to facilitate removal and replacement without damage and arranged so that when sections are removed, instruments, instrument panels, gauge boards and connections need not be disturbed.

Adequate door access shall be provided to the inside of the cover. Where necessary, suitable lighting shall be provided under the cover arranged to produce an adequate lighting level that is compatible with safe working and safe access. All local instrumentation gauges, oil sight glasses, sliprings and the like, local to the turbine-generator are to be visible from the outside of the appearance cover.

4.16 Vibration, Balancing and Critical Speeds

Vibration monitoring equipment shall be provided for each bearing of the main shaft of steam turbine generator set.

Relative shaft displacement vibrations

The steam turbine generator set shall be designed for “Zone A” and has to comply at least with the requirements for vibration magnitude “zone B” stipulated by ISO Specification 7919-2 “Large land based steam turbine generator sets”.

Bearing housing vibration

The steam turbine generator set and its auxiliaries shall be designed for “Zone A” and has to comply at least with the requirements for vibration magnitude “zone B” stipulated by ISO Specification 10816 for the respective group of machinery. .

Individual shafts, discs and couplings shall be dynamically balanced before assembly so that the severity of vibration of the complete unit during operation at any load is within the range specified above.

Any lateral or torsional critical speed of the total combined shaft system including the generator and exciters (if any) shall be at least 15 % above or 10% below the normal range of operating speeds. However, if the critical speed (s) as calculated or revealed on test does not meet the specified separation margin, the response of the rotating assembly to the critical speed(s) in question shall be demonstrated in accordance with API Standards

Care shall be taken in the design of the stationary parts of correcting out-of-balance of the rotors at site without removal of the shafts. Details of the system shall be included in the Bid Proposal.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 Shop Tests

The steam turbine components shall be shop tested and the steam turbine shall be shop assembled to the degree required to ensure proper operating clearances, proper fitup of components for field erection and proper overall operation when the unit is assembled in the field. The Bidder shall describe in the Proposal the shop assembly and testing that will be performed. Refer to **Part 4 Section 3, M-014 "Inspection and Test"**.

Each complete rotor shall be subjected to an overspeed test of 20% above rated speed refer to **Part 4 Section 3, M-014**.

Each rotor shall be statically and dynamically balanced.

a. Steam Turbine and Auxiliaries

All parts subject to steam, oil, hydraulic fluid or water pressure shall be hydrostatically tested refer to **Part 4 Section 3, M-014**.

All equipment shall be tested under simulated working conditions and true working medium (or equivalent) and conducted in accordance with the appropriate ASME Power Test Code.

All valves, controls and their actuators shall be tested to determine their functional integrity. These tests shall include hydrostatic testing of valves for seat leakage in closed position, stem and body leakage in the open position, as well as stroke and timing test of opening and closing devices.

All pumps and motors shall be dynamically balanced together and shall be tested in the Bidder's or subBidder's works in accordance with the latest ASME Power Test Code and Hydraulic Institute Standard for Centrifugal Pumps and IEEE "Test Code for Squirrel Cage Induction motors". Certified pump performance curves shall be submitted to the Owner for approval after tests have been completed.

b. Electrohydraulic Steam Turbine Control System

Shop tests on the steam turbine control system shall be in accordance with **Part 4 Section 3, I-005**.

5.2 Site Tests

Test of steam turbine and auxiliaries shall be in accordance with ANSI unless otherwise specified. The following test, shall be performed:

- a. Centering check
- b. Alignment and leveling check.
- c. Clearance check during assembling.
- d. Tightening records of turbine casing bolts.
- e. Overspeed test.
- f. Bearing inspection.
- g. Oil Flushing.
- h. Load rejection test.
- i. Interlock test.
- j. Load swing test.

5.3 Performance

The Bidder shall conduct performance tests to demonstrate the capacity and efficiencies according to the ASME "Performance Test Codes-Steam Turbines". The Bidder shall furnish a test procedure for testing the equipment in accordance with ANSI/ASME PTC-6. The test procedure shall clearly spell out the instrumentation, test connections, cycle isolation requirements, and any other equipment for the test, and where they must be located in the system to accommodate the required temporary and permanent testing and isolating equipment and instrumentation, such as flow meters, pressure gauges, thermowells, thermocouples etc. and all the required ASME connections on the equipment furnished under this Specification. The Bidder shall liaise with the Owner to record the test data, calculate the required parameters and furnish the Owner with a complete report of the test results. The Bidder shall also arrange to have his authorized representative(s) witness and assist in these tests.

The testing shall be performed at multiple loads as further clarified in **Part 4, Subsection 4.1** specifying general requirements for erection and commissioning.

Further reference is made to Part 4 Subsection 2.6.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

6.1 General

QA/QC of the turbine generator, its subsystems and appurtenances specified here in this Specification shall conform to the requirements of **Part 4, Subsection 2.7**.

6.2 Inspection

The Bidder shall provide inspection to ensure the mechanical accuracy of components, compliance with drawings, and establish the acceptability of all material, parts, and equipment supplied under this Contract. Refer to **Part 4 Section 3, M-014 “Inspection and Test”**.

a. Forgings

Generally refer to **Part 4 Section 3, M-014**.

The turbine and generator rotor forgings shall be ultrasonically tested and inspected in accordance with ASTM A418-77. The scanning shall be continuous over the whole surface of the rotor, and the rotor surface for testing shall be smooth enough to obtain an accurate indication.

The entire surface of the rotor forgings shall also be magnetic particle inspected.

After machining, rotor end winding retaining rings shall be radiographed, ultrasonically tested and further examined for cracks using the dye penetrant method.

All other steel forgings shall be inspected by magnetic particle inspection at the regions of fillets and change of section. Any flaws which cannot be removed during final machining shall be repaired using a repair procedure approved by the Owner.

All forgings shall be demagnetized after magnetic particle inspection.

b. Castings

Generally refer to **Part 4 Section 3, M-014**.

Test specimens shall be tested in accordance with ASTM E8-81.

Complete visual and dimensional checks shall be conducted.

All casing surfaces shall be examined by magnetic particle inspection in accordance with ASTM E709 with basis for acceptance judged by reference to ASTM E125.

All defects beyond the acceptance limits of the above standard requiring weld repair shall be reported to the Owner, and a repair procedure submitted for approval prior to any repairs being carried out. Weld repair shall be mapped and submitted to the Owner with the test reports.

Following major repairs, all castings shall be fully annealed. Smaller repairs must be followed by stress relieving. All completed repairs shall be fully radiographed.

All weld preparation areas on the casings shall be fully radiographed.

c. Welds

All welds manufactured as parts for the turbine generator shall be fabricated only with materials meeting the requirements of the ASME Boiler Code, Section II, Material Specifications.

The Bidder shall have all shop welded procedures, welding equipment, and operators qualified in accordance with the ASME Boiler Code, Section IX, Welding Qualifications.

d. Miscellaneous

All fixed and moving blades and shrouding shall be magnetic particle inspected prior to installation. Spot check type dye penetrant inspection shall be performed on these parts after overspeed tests.

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall be responsible for the complete design, supply and installation of all instrumentation and control systems and devices listed and described in this technical specification.

For the functional requirements in view of turbine start up, supervisory and control, etc. of the Electrohydraulic Steam Turbine Control System, refer to **Part 4 Section 3, I-100 “Plant Control System”**

7.2 Design Requirements

The design of instrumentation and control systems, panels and devices shall be in accordance with the requirements detailed in **Part 4, Section 3:**

- I-001 Instrumentation and Controls
- I-003 Local Mounted Instrument Panels
- I-002 Local Control Panels
- I-005 Distributed Digital Control System
- I-100 Plant Control Systems
- **I-300 Block Control Desk and Panels (Not applicable)**

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7.3 Gland Steam Sealing System

The Bidder shall furnish and install all local instrumentation and controls.

The Bidder shall supply and install remote switches to control all motorized valves and the gland steam condenser, air exhausters, and all interconnecting cables and wiring.

The desuperheating of sealing steam for the glands, and elsewhere as appropriate, shall be fully automatic under all operating conditions. The gland steam temperature downstream of any desuperheater shall be monitored locally and remotely and shall also include and abnormal temperature alarm.

Local indication of pressure in the gland sealing steam supply and exhaust manifold shall be provided, appropriately located in the vicinity of the pressure reducing control valves to facilitate local manual adjustment to the control if

this should be necessary. Means shall also be included to give alarms when these pressures are abnormal.

7.4 Water Induction Prevention

The Bidder shall furnish complete logic describing the operation and interlocking of all the local instrumentation and control and inline devices and accessories for and associated with the prevention of water induction into the turbine. The interlock logic diagram shall be in accordance with the requirements and intent of ASME Standard No. TDP-1-1 985 as a minimum.

7.5 Turbine Instruments

All supervisory instruments furnished with the turbine generator shall be provided with alarm contacts, rated 0.5 amperes inductive at 220 V DC.

In addition to such instruments and devices which are specified for the steam turbine generator set and its appurtenances, the following instruments and devices shall be furnished for the turbine if required

7.5.1 Pressure Gauges

- a. The following pressure gauges, and such others required for turbine operation, shall be furnished
 - Hydraulic gear pressure gauges
 - Oil pressure gauges (each bearing)
 - Gland system pressure gauges
- b. The gauges shall be mounted on the turbine or on turbine piping, or flush-mounted on a suitable panel mounted on or near the turbine, as approved by the Owner. Control panels furnished with the equipment shall be completely piped and wired in the shop.
- c. Pressure gauge cases shall be dustproof, not less than 4-1/2 in. In size except for special applications where smaller sizes are more practical, and made of material and finished as approved. Gauges shall be suitable for the services intended.

7.5.2 Thermometers

- a. The following thermometers and such others required for turbine operation shall be furnished

- Oil return from each bearing
 - Oil inlet on oil coolers
 - Oil outlet on oil coolers
 - Oil outlet on thrust bearing
 - Exhaust hood
- b. These shall be dustproof dial-type thermometers of suitable materials. Each thermometer shall be provided with a separate socket.

7.5.3 Alarm and Trip Devices

The following alarm devices shall be furnished:

- a. All alarm and trip contacts associated with the turbine and which are required for safe turbine (and auxiliary) operation shall be provided (e.g., low oil pressure, low electric-hydraulic (EH) pressure, low vacuum trip, thrust bearing failure, etc.)
- b. Alarm devices shall have contacts rated 0.5 ampere inductive at 220 V DC
- c. Necessary contacts on throttle valves, governor valves, etc., as required for interlocking devices, alarms

7.5.4 Instrument Piping

Piping for connecting instruments shall be furnished as herein specified.

7.5.5 Pressure Transmitters

- a. Bidder shall furnish the following electronic-type transmitters (smart type) and appropriate power supply
 - Steam chest pressure (if required)
 - First stage pressure
 - Reheat bowl pressure
 - Steam seal pressure
 - Lube oil pressure
 - Hydraulic oil pressure
 - Gland exhaust pressure

- LP Turbine exhaust pressure
- b. The above list is in addition to the transmitters which are normally provided in the turbine manufacturer's standard scope of supply

7.5.6 Thermocouples

Bidder shall provide chromel-constantan or N-type (Nicrosil/Nisil) thermocouples (and thermowells) for each of the following measurements, to be used for input to the steam turbine safety and control system and the DCS for operator interface monitoring, alarms and logging:

- a. Turbine metal temperatures
- b. Water detection temperatures, which are required in the turbines.
- c. Exhaust hood temperature(s)
- d. Steam-seal header temperatures
- e. Thrust and journal bearing oil drain temperatures
- f. Thrust and journal bearing metal drain temperatures
- g. Lube oil to and from cooler temperatures
- h. LP turbine inlet temperatures

7.5.7 Operator Interface Monitoring and Primary Elements

Bidder shall provide signals for operator interface monitoring for the following items:

- a. Thermal stress monitoring
- b. Steam-seal header temperatures
- c. Journal and thrust bearing metal temperatures
- d. Lube oil to and from cooler temperature
- e. All turbine supervisory inputs

7.5.8 Turbine Supervisory Instruments

Bidder shall provide all turbine supervisory instruments that are necessary to assure safe operation, startup, and shutdown of the turbine including, but not necessary limited to, the following:

- a. Vibration on each bearing
- b. Differential expansion
- c. Generator Power Output at terminals
- d. Shell expansion.
- e. Turbine speed, zero speed, and acceleration
- f. Governor or control valve position
- g. Vibration phase angle

Bidder shall provide necessary 4-20 DC mA output signals from the supervisory instrumentation for inputs to TSI cabinet.

7.5.9 Resistance Temperature Detectors

Bidder shall provide three-wire, RTDs with thermowells for operator interface monitoring, alarm, and logging.

- a. Hydrogen to each cooler
- b. Hydrogen from each cooler
- c. Hydrogen common cold gas
- d. Generator stator coil water or hydrogen
- e. Exciter air temperature (if required)

7.5.10 Thermocouples, RTDs, and Transducers

All thermocouples, RTDs, and transducers shall be wired to conveniently located terminal blocks for receiving electrical connections. All necessary piping from the point of measurement to the transducers shall be provided by the Bidder. Transducers shall provide 4 to 20 miliampere output signal.

7.6 Equipment Sound Levels

The warranted maximum A-weighted sound levels under any operating condition measured one meter from any equipment supplied shall meet, when measured in accordance with ANSI S1.13, not exceed 85 dBA, refer also to **Part 4 Subsection 2.9 and 2.16**. This equipment shall include but not be limited to:

- a. Turbine casings

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- b. Couplings
- c. Generator casings or generator noise enclosure if used.
- d. Exciter
- e. Lubrication system
- f. EH control (EHC) high-pressure unit
- g. Gland steam sealing system
- h. Cooling system
- i. Valves
- j. Motors
- k. Others, if any, as specified in the **Part 4** Specification

7.7 Automation Control Equipment

All necessary equipment shall be provided as required for automation of the turbine-generator unit. This shall include those items of equipment, as recommended by manufacturer, that will allow complete turbine startup, operation, load change, and shutdown by DCS.

8. ELECTRICAL SPECIFICATION

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Steam Turbine.

For general electrical requirements, see **Part 4 Section 3, E-001**.

8.2 Generators

For detailed electrical requirements, see **Part 4 Section 3, E-101**.

8.3 Motors

For detailed electrical requirements, see **Part 4 Section 3, E-210** for Low Voltage Motors.

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